Walchand College of Engineering, Sangli
Computer Science & Engineering
Third Year
Course: Computer Algorithm Lab

Assignment No. 10

Complexity classes and Approximation algorithms:

1) In defense systems or sensor networks, placing radars/detectors such that none interfere with each other mimics the n-Queen problem.

Implement the n × n Queen problem using backtracking.
Extend the solution for n = 8 and display all possible solutions.
Analyze the time complexity and discuss why it's an NP problem.

2) In wireless networks or exam scheduling, we must assign frequencies/time slots so that no two adjacent nodes (or exams) share the same color.

Implement Graph Colouring using Backtracking.
Input: Graph adjacency matrix.
Output: Minimum number of colors required and coloring assignment.
Verify that adjacent vertices do not share the same color

3) In logistics and delivery networks, a courier must visit every depot exactly once and return to the starting point without repeating any location — this mirrors the Hamiltonian Cycle problem.

Implement the Hamiltonian Cycle problem using backtracking.
Input: Graph adjacency matrix representing depots and the roads connecting them.
Output: Display the Hamiltonian cycle if one exists, or report that none exists.
Analyze the time complexity and discuss why it belongs to the NP-complete class.

4) In network security, placing monitoring devices on the minimum number of routers such that every communication link is observed by at least one device mimics the Vertex Cover problem.

Implement an Approximation Algorithm for the Vertex Cover problem.
Input: Graph adjacency matrix/list representing routers and communication links.
Output: The approximate vertex cover set and its size.
Compare the approximate solution's size against the optimal (where feasible for small graphs), and derive/discuss the algorithm's approximation ratio (2-approximation).